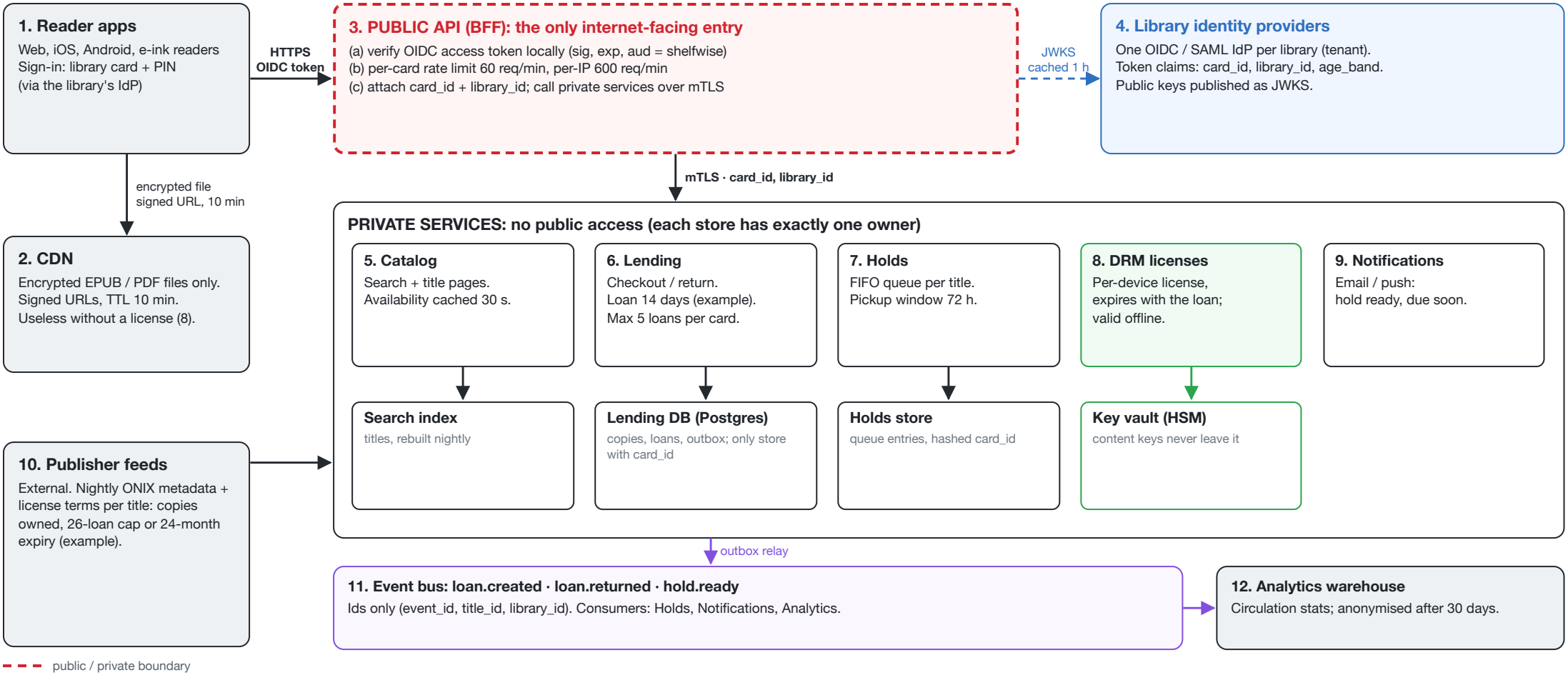
Shelfwise (fictional): E-book Lending Platform

Target architecture: public entry, private services and the stores they own, events, failure handling and SLOs



13. Failure modes

Search index down:
serve cached results for 5 min, then "search unavailable"; checkout by title_id still works.
Lending DB down:
FAIL CLOSED on checkout / return (never oversell copies); open books keep working (licenses are offline).
DRM license service down:
existing licenses unaffected; new checkouts retry for 2 min, then fail with Retry-After.
Event bus down:
outbox keeps events; notifications delayed, never lost.
Library IdP down:
signed-in readers continue until token expiry (1 h); new sign-ins fail for that library only.

14. SLOs and observability

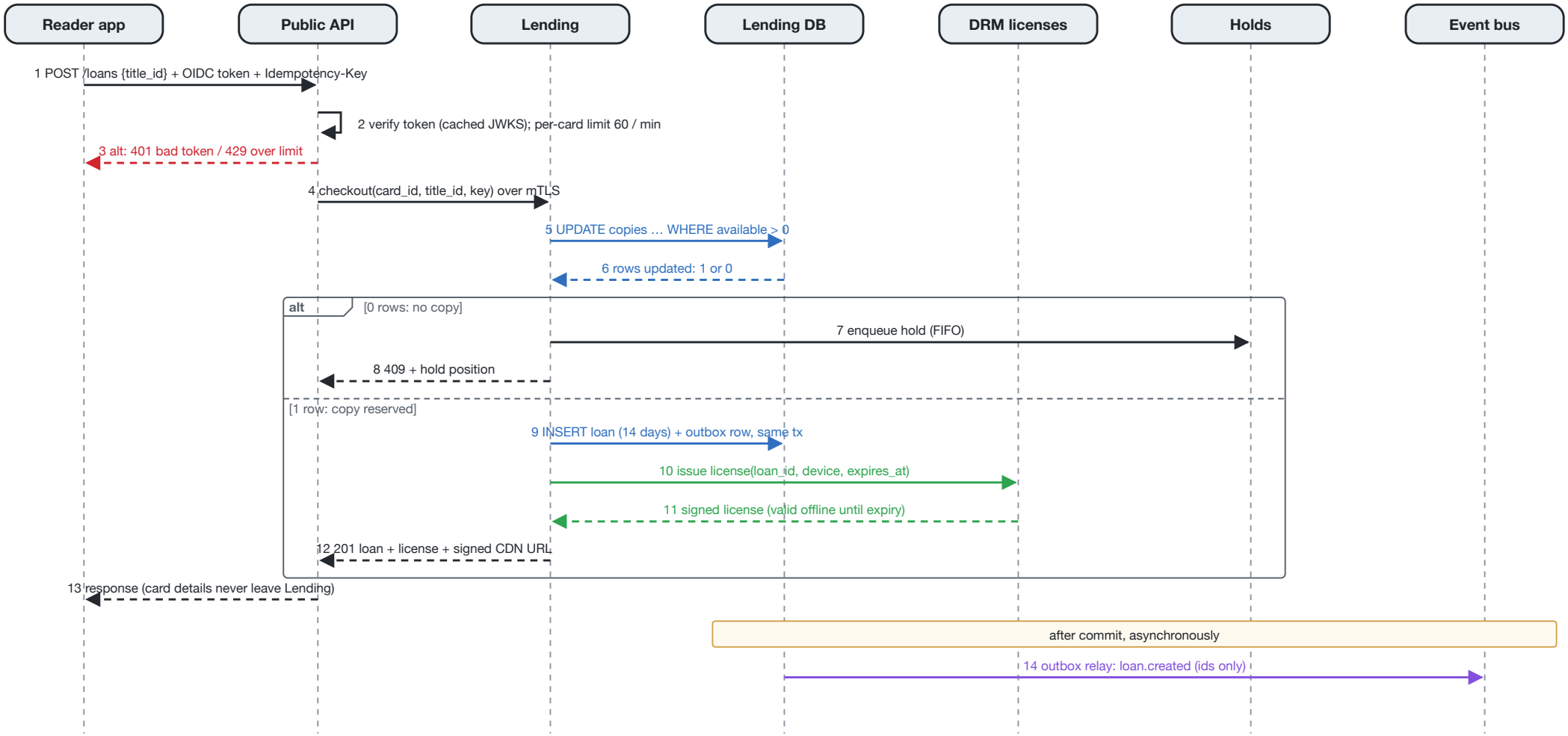
Checkout p99 < 400 ms; availability 99.9 % / month.
Metrics: checkouts / min, hold-queue length, license errors, outbox lag.
Audit per loan action: hashed card_id, title_id, library_id, action, request_id.
Alerts: copies_available < 0 (must never happen), license error rate > 1 %, outbox lag > 60 s.

15. Data contracts

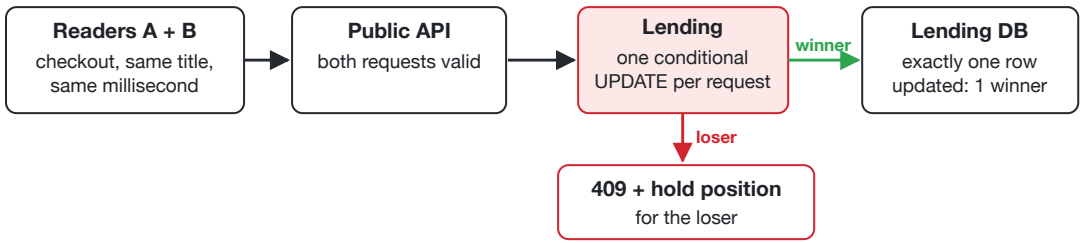
Event envelope: event_id, type, occurred_at, title_id, library_id, schema_version.
No card_id, name or email in any event.
Delivery: at-least-once; consumers dedupe on event_id (kept 7 days).
Retention on the bus: 7 days.
Publisher feed: signed ONIX files; rejected if the signature or schema fails, previous terms stay active.

Shelfwise (fictional): E-book Lending Platform

Flow A: borrow an e-book, the last-copy race, and key decisions



Incident class: two readers check out the last copy at once



UPDATE copies SET available = available - 1 WHERE title_id = ? AND available > 0, in the loan transaction.
Idempotency key per checkout (kept 24 h), so a client retry never creates a second loan.

Alert if copies_available < 0 or active loans > licensed copies; audit every 409.
Fixes the incident class: overselling licensed copies, which breaches the publisher contract.

Key decisions

1. One public entry (BFF); services unreachable from the internet
2. Per-library IdPs; API verifies tokens locally with cached JWKS
3. Copies reserved by one conditional UPDATE, no distributed lock
4. Idempotency key on every checkout, kept 24 h
5. Licenses valid offline: a DRM outage is not a reading outage
6. card_id lives only in the Lending DB; events carry ids only
7. Transactional outbox for every loan event
8. Analytics anonymised after 30 days (example)